encompasses a user-friendly mobile application, providing individuals with real-time access to their health information.

The application described here delivers personalised feedback, health recommendations, and alerts based on the analyses performed by R. Moreover, healthcare providers can remotely monitor patients' health status and intervene promptly if anomalies are detected.

To ensure the security and privacy of health data, the system incorporates robust encryption and authentication mechanisms. The cloud-based infrastructure, leveraging R, ensures secure data storage and compliance with healthcare data protection regulations.

This proposed IoT connected health monitoring system exemplifies the substantial contribution R can make to the evolving field of digital healthcare. The objective of this system is to formulate an IoT based healthcare application that can predict the possibility of a heart attack through smart wearables, using generalised linear models (GLMs), random forests and decision trees in R.

The problem statement

The diagnosis of heart disease is a difficult task, but automation leads to better predictions about a patient's heart condition. Data relating to resting blood pressure, cholesterol, age, sex, type of chest pain, fasting blood sugar, ST depression, and exercise-induced angina can all help to predict the likelihood of having a heart attack. Decision trees, random forests and GLMs can be trained on the given dataset and be used to view the predicted classes, where 0 equals less chances and 1 equals more chances of a heart attack.

Setting up the health monitoring system

- Import the packages Rplot, RColorBrewer, Rattle and randomForest.
- Download and read the dataset from Kaggle at *https://www.kaggle.com/ datasets/nareshbhat/health-care-data-set-on-heart-attack-possibility.*
- View the statistics of the variables in the dataset using the function 'summary'.
- Analyse the data, specific to resting blood pressure.
- Using the 'cor' function, find the correlation between resting blood pressure and age.
- Construct a logistic regression model using a GLM and view the output plots.
- Encode the target values into categorical values.
- Split the dataset into training and testing data in the ratio 70:30.
- Construct a decision tree model.
- Target variable is categorised based on resting blood pressure, serum cholesterol and maximum heart rate achieved.
- Plot the decision tree and view the output.
- Devise a random forest model based on the relationship between resting blood pressure, old peak

Installing the necessary packages

In [1]:
```
install.packages("rpart.plot")
install.packages("rattle")
install.packages("randomForest")
```

```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

also installing the dependencies 'bitops', 'XML'


Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

Loading all the necessary Libraries

In [2]:
```
library(rpart) #used for building classification and regression trees.
library(rpart.plot)
library(RColorBrewer) # help you choose sensible colour schemes for figure
library(rattle) # provides a collection of utilities functions for a data
library(randomForest) #has the function randomForest() which is used to c
```

```
Loading required package: tibble

Loading required package: bitops

Rattle: A free graphical interface for data science with R.
Version 5.5.1 Copyright (c) 2006-2021 Togaware Pty Ltd.
Type 'rattle()' to shake, rattle, and roll your data.

randomForest 4.7-1

Type rfNews() to see new features/changes/bug fixes.


Attaching package: 'randomForest'


The following object is masked from 'package:rattle':

    importance
```

Figure 1: Installing and loading libraries